MTH636 – Game Theory

Lecture 22: Extensive Form Games with Imperfect Information

1 Motivation: Imperfect Information in Extensive Form

So far we have studied extensive form games with *perfect information*, where each player observes the full history of play before making a decision. In many real-world situations, however, a player may not know which node in the game tree they currently occupy. We now introduce the framework of **extensive form games with imperfect information** (and possible chance moves).

1.1 Illustrative Examples

Battle of the Sexes – two versions. Consider the standard Battle of the Sexes scenario with payoffs $(2,1)$ for (F,F), $(0,0)$ for (F,D), $(0,0)$ for (D,F), and $(1,2)$ for (D,D).

- **Left game (Man moves first):** The root is labelled M (Man). Man chooses F or D. Woman (W) observes Man's choice and responds. Since Woman can see Man's move, the two Woman nodes are in *separate* information sets – this is a game of **perfect information**.
- **Right game (Woman moves first):** The root is labelled W (Woman). Woman chooses F or D. Man (M) does *not* observe Woman's choice, so his two decision nodes are connected by a dashed oval, indicating a single information set – this is a game of **imperfect information**.

In both cases the terminal payoffs are $(2,1)$, $(0,0)$, $(0,0)$, $(1,2)$ at the four leaves.

A richer example with Nature. Consider a game tree rooted at n_0 (Player I). Player I chooses A, B, or C.

- After A: reach n_1. After B: reach n_2. Player II has information set $\{n_1, n_2\}$ (dashed oval – she cannot distinguish n_1 from n_2). From this information set she plays ℓ or g. Outcomes: $(4,1)$, $(7,2)$ from n_1, and $(6,3)$, $(4,0)$ from n_2.
- After C: reach n_3. Nature (O) moves, playing $\frac{1}{4}$ toward n_4 and $\frac{3}{4}$ toward n_5. At n_4, Player II chooses L or R. At n_5, Player I chooses a or b.
- There is also an information set $\{n_1, n_2, n_5\}$ visible in the diagram.

2 Formal Definition

Definition: Extensive Form Game with Imperfect Information

A game in extensive form (with imperfect information and chance moves) is a tuple

$$\Gamma = \left\langle N,\, V,\, E,\, n_0,\, (V_i)_{i\in N\cup\{0\}},\, \mathcal{H} = (H_i)_{i\in N},\, \mathcal{A} = (A(h))_{h\in\bigcup_i H_i},\, (p_n)_{n\in V_0},\, (u_i)_{i\in N} \right\rangle$$

where the components are defined as follows.

2.1 Components of the Tuple

- N: the set of **players**.
- (V, E, n_0): a **directed rooted tree**.
- $(V_i)_{i\in N\cup\{0\}}$: a **partition** of the set of non-leaf nodes among the agents and nature. V_0 is the set of *nature's nodes* (chance nodes); V_i is the set of nodes at which player i moves.
- $\mathcal{H} = (H_i)_{i\in N}$: H_i is a **partition** of V_i. Each set in the partition is called an **information set**. Each $h \in H_i$ must satisfy:
  (i) Each node in h has the **same number of outgoing edges**.
  (ii) Every path in (V, E, n_0) intersects h **at most once**.
- $\mathcal{A} = (A(h))_{h\in\bigcup_{i=1}^{n} H_i}$: for each information set $h \in \bigcup_i H_i$, $A(h)$ is a **partition of the edges** coming out from the nodes in h, such that for all $a \in A(h)$, a contains exactly one edge outgoing from n for all $n \in h$.

  A set $a \in A(h)$ is called an **action at** h.
- $(p_n)_{n\in V_0}$: p_n denotes a **probability distribution** over the edges outgoing from n (for each chance node n).
- $(u_i)_{i\in N}$: the **utility functions** defined at the leaf nodes.

3 Information Sets: Validity Conditions

3.1 Invalid vs. Valid Game Trees

The two conditions on information sets – (i) equal number of outgoing edges at every node in h, and (ii) every root-to-leaf path crosses h at most once – are essential for the game to be well-defined.

Example of an invalid game. Consider a two-player tree: Player I at n_0 chooses L or R, leading to n_1 or n_2. From n_1, Player II moves and then the game continues to a subtree containing n_3 and n_4. From n_2, the game continues to n_4. If n_3 and n_4 are in the same information set of Player II, then there exists a path from the root that passes

through this information set *more than once*, violating condition (ii). This is **not a valid extensive form game**.

Note

The annotated note in the lecture reads: "Not a valid extensive form game."

Example of a valid game (not satisfying perfect recall). Consider Player I at n_0 choosing L or R to reach n_1 or n_2. Player II has information set $\{n_1, n_2\}$ (she cannot distinguish them). From each she plays a, b, d (or similar). A second Player II information set groups some descendants together. Every path from the root crosses each information set at most once, so this is a **valid** extensive form game – however it **does not satisfy perfect recall** (noted in red in the lecture).

4 Perfect Recall

Definition: Perfect Recall

An extensive form game satisfies **perfect recall** for player i if for all information sets $h \in H_i$ and for all $u, y \in h$, an action of player i is on the path from the root to u if and only if it is on the path from the root to y.

In other words, perfect recall requires that whenever two nodes belong to the same information set of player i, the player must have taken exactly the same sequence of actions to reach either node – she has not "forgotten" any earlier choices she made.

4.1 Example of a Violation of Perfect Recall

Consider the following game tree:

- Player I (n_0) chooses L or R.
- After L: Player II (n_1) chooses ℓ or r, reaching n_3 or n_4.
- After R: reach n_2, then Player II (n_2) chooses ℓ or r, reaching n_4 or n_5.
- Nodes n_3, n_4, n_5 are grouped into a single Player II information set.
- Player II chooses α or β from this information set.

In this game, the path to n_4 can arise via two different routes: $n_0 \xrightarrow{L} n_1 \xrightarrow{r} n_4$ (Player II moved at n_1) or $n_0 \xrightarrow{R} n_2 \xrightarrow{\ell} n_4$ (Player II moved at n_2). In both cases Player II ends up in the information set $\{n_3, n_4, n_5\}$, but whether she earlier chose r (at n_1) or ℓ (at n_2) differs between the two paths. Thus Player II has "forgotten" her own earlier move.

Key Insight: Violation of Perfect Recall

This game **violates perfect recall** for Player II (noted in red in the lecture). Perfect recall is violated whenever a player cannot remember their own past actions when entering an information set.

Note

Perfect recall is a standard assumption in most game-theoretic models. When it holds for all players, Kuhn's theorem guarantees that mixed strategies and behaviour strategies are *outcome-equivalent*.